

Batch

1. Worked example: one row (cpdino on Positiv_16)

n_true 295 | n_pred 284 | count_diff -11 | count_error -0.037

TP 229 | FP 55 | FN 66 | precision 0.806 | recall 0.776 | F1@0.5 0.791 | F1@0.75 0.608

gt_median_diam_px 45.8 | time_s 3.58

IMES annotated **295** cells, and the model found **284**.

- **229** of the 284 overlap a real cell well enough → **TP** (correct)
- **55** don't match any real cell → **FP** (false)
- **66** real cells were not found → **FN** (missed)

From these:

- **precision** = $229 / 284 = \mathbf{0.806}$: 81% of what it found is real.
- **recall** = $229 / 295 = \mathbf{0.776}$: it found 78% of the real cells.
- **F1@0.5** = $2 \cdot 229 / (2 \cdot 229 + 55 + 66) = \mathbf{0.791}$: one score combining both.
- **F1@0.75** = **0.608**: the same score, but with a stricter overlap rule, so it measures outline accuracy.
- **count_diff** = $284 - 295 = \mathbf{-11 \text{ cells}}$, and **count_error** = $-11 / 295 = \mathbf{-3.7\%}$.
- **gt_median_diam_px** = **45.8**: the typical annotated cell in this image is about 46 pixels wide.

How “matching” works: overlap is measured with **IoU** (shared area ÷ combined area; 1 = identical). Each annotated cell can be matched to at most one prediction. At @0.5 a pair counts if IoU ≥ 0.5 ; at @0.75 it needs ≥ 0.75 . So if a model cuts one cell into two halves, one half is a TP and the other an FP. That's why splitting cells is punished.

2. per_image.csv: the 17 columns

Column	Meaning	How to read it
model	which Cellpose model	
image	image name	
series	experiment batch: CLD-AIM-04, PAD11a, PAD11c	later used to keep train and test images apart
condition	differentiated (Positiv, Diff) or control (Negativ, Ctrl)	control wells were not turned into fat cells
n_true	annotated cells	the “correct answer”
n_pred	cells the model found	
count_diff	$n_pred - n_true$	+ too many, – too few
count_error	$count_diff \div n_true$	0.149 = +14.9%, 5.2 = +520%
TP@0.5	correctly found cells	higher is better
FP@0.5	false detections	lower is better
FN@0.5	missed cells	lower is better
precision@0.5	$TP \div n_pred$	low → the model over-detects or splits cells
recall@0.5	$TP \div n_true$	low → the model misses cells
F1@0.5	$2TP \div (2TP + FP + FN)$	headline: 1 = perfect, 0 = useless
F1@0.75	the same with the stricter overlap rule	outline quality , which matters for cell size
gt_median_diam_px	typical annotated cell diameter in pixels	used for the --diameter test
time_s	seconds for this image	compare only within one run

3. summary.csv: the 17 columns

Each row is one model × one group: control, differentiated, or all (both together).

Column	Meaning
n_images	number of images in this group
n_true_total / n_pred_total	all annotated / all found cells added up
F1@0.5_mean	average F1 over the images (every image counts equally)
F1@0.5_std	spread between images. Small = consistent, large = good on some images, bad on others
F1@0.5_min / _max	worst / best single image
pooled_F1@0.5	F1 from all cells together (sum of all TP, FP, FN), so every cell counts equally
F1@0.75_mean	average outline quality
precision@0.5_mean / recall@0.5_mean	averages over images
abs_count_error_median	typical size of the count error on one image. Median, so a few disastrous images don't distort it
pooled_count_error	(all found – all annotated) ÷ all annotated: is the total count right?
time_s_mean	average seconds per image
peak_gpu_gb_max	highest GPU memory used

Mean vs pooled: if the two are close, the model behaves evenly across images. If pooled is much lower, a few images with huge numbers of false cells are dragging it down.

4. The results (read the differentiated rows: 25 images, 8,196 annotated cells)

	cpsam_v2	cpsam	cpdino	cpdino-vitb
F1@0.5 mean $\pm$ std	0.43 $\pm$ 0.26	0.47 $\pm$ 0.27	0.62 $\pm$ 0.24	0.65 $\pm$ 0.14
F1@0.5 worst / best image	0.02 / 0.74	0.01 / 0.76	0.07 / 0.80	0.30 / 0.80
Pooled F1@0.5	0.28	0.29	0.51	0.65
F1@0.75 (outlines)	0.28	0.28	0.38	0.37
Precision / recall	0.42 / 0.48	0.45 / 0.50	0.60 / 0.66	0.69 / 0.63
Cells found (annotated 8,196)	19,798	19,301	12,981	7,474
Pooled count error	+142%	+136%	+58%	-9%
Median count error per image	18%	20%	11%	13%
Time / GPU memory	3.7 s / 1.6 GB	3.7 s / 1.6 GB	3.6 s / 1.3 GB	1.7 s / 0.7 GB

What the numbers tell us:

- **The Cellpose-SAM models (cpsam_v2, cpsam) find about 2.4× too many cells.** Precision is low (0.42–0.45): they split cells, and on Positiv_21–26 they count every lipid droplet as a cell (up to 3,332 “cells” where 394 are annotated).
- **cpdino is the best on typical images.** On 19 of the 25 images it scores 0.65–0.80 with a typical count error of about 11%, and it draws the best outlines. But on Positiv_21–26 it still over-segments (F1 0.07–0.48), which pulls its pooled F1 down to 0.51.
- **cpdino-vitb is the steadiest.** Its worst image still scores 0.30 (the others go down to 0.01–0.07). Mean and pooled F1 are equal (0.65), and its **total** count is almost right (–9%). It’s cautious on the difficult images: what it finds is mostly correct, but it misses some cells there. It’s also twice as fast and uses half the memory.
- **All 4 models fail on control wells.** They find 135–486 objects where only 12 are annotated in total. The control rows’ count errors (13–25, i.e. +1300% to +2500%) are huge only because the annotated numbers are so small.

- **New insight from gt_median_diam_px:** cells are about 32–46 px wide in every image, **including Positiv_21–26 (33–40 px)**. So those images fail not because the cells are a different size but because of the large droplets inside them. The --diameter resizing test may help less than hoped. It's still worth one try, but fine-tuning is the more likely fix.

5. Final verdict

Rank	Model	Verdict
1	cpdino (Cellpose + DINOv3 ViT-L)	Most accurate where images look normal: best F1 on typical images, best outlines, smallest typical count error. Reference model for zero-shot comparisons.
2	cpdino-vitb (Cellpose + DINOv3 ViT-B)	Most robust and efficient: best mean and pooled F1, never collapses, total count nearly right, 2× faster, 0.7 GB. First choice for fine-tuning on the 8 GB GPU.
3	cpsam (original Cellpose-SAM)	Clearly behind the DINO models; over-segments.
4	cpsam_v2 (current Cellpose default)	Last. Drop it as the default reference.

What's solid: the DINO-backbone models beat Cellpose-SAM on 24–25 of 25 differentiated images ($p < 0.001$), and two independent runs gave identical numbers.

What isn't settled:

- Which DINO model is “best” depends on whether you value accuracy on typical images (cpdino) or robustness (cpdino-vitb).
- No model is ready for IMES as-is, because of the droplet problem on mature cells and the over-detection in control wells.

Limits:

- 1 of 4 folders tested (porcine brightfield only).
- Default settings, nothing tuned.
- The annotation rule (“only fat cells”) isn't confirmed yet.

One sentence for the PDF or meeting:

“On 28 porcine brightfield images, Cellpose with a DINOv3 backbone clearly beats Cellpose-SAM zero-shot (mean F1@0.5 0.62–0.65 vs 0.43–0.47; better on 24 of 25 images). cpdino is the most accurate on typical images and cpdino-vitb the most robust and efficient, but all models still count large lipid droplets as cells and over-detect in control wells, which motivates fine-tuning and a fat-cell selection step.”

What are differentiated images ?

“Differentiated” is a label **our script gives each image based on its filename**. It describes how the cells in that well were treated.

The biology:

- IMES starts with **stem cells**, which can still become different cell types.
- In some wells they add a special growth medium that pushes the stem cells to **turn into fat cells**. This process is called **differentiation**. Those cells fill up with lipid droplets: the “dots” you see in the images.
- Other wells get **no** such treatment. Those cells stay as plain stem cells. These are **control** wells, used for comparison.

How the script decides, from the filename:

Filename contains	Label	Meaning	Porcine BF images
Positiv or _Diff	differentiated	cells were pushed to become fat cells → many fat cells	25 (23 Positiv + 2 Diff)
Negativ or _Ctrl	control	no treatment → mostly stem cells, very few fat cells	3 (1 Negativ + 2 Ctrl)

You can see the difference in the overlays from earlier:

- **Positiv_22 (differentiated)**: the image is full of droplet-filled fat cells, about 394 annotated.
- **Negativ_2 (control)**: a dense layer of plain stem cells, with only 5 annotated fat cells.

Why the results are split this way: in control images there are only 2–5 annotated cells. A model that finds 50 objects there gets a terrible score, even though those

objects may be real stem cells that simply weren't annotated. If we averaged control and differentiated images together, these 3 images would distort the result. So the main comparison uses the **25 differentiated images**, and the control images are reported separately. They answer a different question: "does the model wrongly find fat cells where there are almost none?"

One assumption to confirm with Nicole: reading Positiv/Diff as "differentiation medium added" and Negativ/Ctrl as "no medium" is my interpretation of the IMES naming. The images fit it, but it isn't written in the README.

```
=== summary per model and condition (mean +- std over images; pooled = all cells together)
model      condition      n      F1@0.5  pooled  F1@0.75  cells pred/GT med |count err|  s/img
cpsam_v2   control              3    0.09 ±0.024  0.082    0.039    135/12          13.2  3.62
cpsam_v2   differentiated      25   0.434 ±0.258  0.275    0.276   19798/8196       0.18  3.67
cpsam_v2   all                  28   0.397 ±0.266  0.274    0.251   19933/8208       0.305  3.66
cpsam      control              3    0.188 ±0.184  0.105    0.054    141/12          14.8  3.54
cpsam      differentiated      25   0.465 ±0.268  0.291    0.279   19301/8196       0.197  3.7
cpsam      all                  28   0.435 ±0.272  0.29     0.255   19442/8208       0.209  3.68
cpdino     control              3    0.06 ±0.056  0.058    0.037    229/12          16.4  3.39
cpdino     differentiated      25   0.617 ±0.237  0.506    0.38     12981/8196       0.106  3.57
cpdino     all                  28   0.557 ±0.284  0.501    0.344   13210/8208       0.115  3.55
cpdino-vitb control              3    0.063 ±0.059  0.024    0.063     486/12          25.5  1.5
cpdino-vitb differentiated      25   0.65 ±0.135  0.653    0.37     7474/8196       0.125  1.68
cpdino-vitb all                  28   0.587 ±0.225  0.634    0.337    7960/8208       0.139  1.66
```

Each model gets three rows: **control** (3 images), **differentiated** (25 images) and **all** (both together, 28 images).

Column	Meaning	Example (cpdino, differentiated)
n	number of images in the row	25
F1@0.5	average F1 over the images ± spread between images	0.617 ± 0.237
pooled	F1 computed from all cells together, so every cell counts equally	0.506
F1@0.75	average outline quality (stricter overlap rule)	0.38
cells pred/GT	all cells found / all cells annotated	12,981 / 8,196, about 1.6× too many

Column	Meaning	Example (cpdino, differentiated)
med count err	typical count error on one image, as a fraction	0.106 = about 11% off
s/img	average seconds per image	3.57

How to read the three row types:

- **differentiated: the row to use for the verdict.** These are the 25 fat-cell images. cpdino-vitb leads on mean and pooled F1 (0.65 / 0.653). cpdino has the best outlines (0.38) and the smallest typical count error (0.106).
- **control:** only 12 annotated cells in total across 3 images. All models find far more (135–486). So F1 is very low and the count error is huge: 13.2 means +1320%, and 25.5 means +2550%. These rows only show that **no model is ready for control wells**. They don't rank the models.
- **all:** both mixed together, so the 3 control images pull the scores down a little. Not needed for the comparison.

Two things the table makes easy to see:

- **The gap between F1@0.5 and pooled shows how evenly a model behaves.** cpdino-vitb: 0.65 vs 0.653, very even. cpsam_v2: 0.434 vs 0.275, because a few images with thousands of false “cells” ruin its overall total.
- **The cells pred/GT column shows the over-counting directly.** The two Cellpose-SAM models find about 19,800 cells where 8,196 are annotated (2.4×). cpdino-vitb finds 7,474, closest to the truth.